BTS – VI – 04.16-0316 Reg. No. | | | | | | | | |

B

B.Tech. Degree VI Semester Examination April 2016

ME 1602 MACHINE DESIGN I

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

(*Use of approved design data hand book is permitted*)

PART A

(Answer ***ALL*** questions)

($8 \times 5 = 40$)

I. (a) What is the difference between failure due to static load and fatigue failure?

(b) What are thermal stresses and creep? How are they considered in the design of machine elements?

(c) What is self locking of power screw? What is the condition of self locking?

(d) Give at least three practical applications of couplings.

(e) What are the various modes in which a riveted joint may fail?

(f) What is the objective of nipping of leaf spring?

(g) Discuss the advantages of welded joints in comparison to riveted joints.

(h) Explain equivalent torsional moment and equivalent bending moment in shafts.

PART B

($4 \times 15 = 60$)

II. A cylinder shaft made of steel of yield strength 700 M Pa is subjected to static loads consisting of bending moment 20 kN-m and a torsional moment 40 kN-m. Determine the diameter of the shaft using maximum principal stress theory, maximum shear stress theory and distortion energy theory, assuming a factor of safety of 2. Take E=210 GPa and Poison's ratio = 0.25.

OR

III. A rod of circular cross-section is subjected to an alternating tensile force varying from 25 kN to 75 kN. Determine the diameter of rod according to Gerber methods, Goodman method and Soderberg method using the following material properties.

Take ultimate strength as 1100 MPa, Yield strength as 600 MPa, Factor of safety as 2 and neglect other correction factors.

IV. A mild steel shaft has to transmit 80 kW at 250 rpm. The abllowable shear sress in the shaft material is limited to 50 MPa and crushing stress limited to 100 MPa. The angle of twist of shaft is not to exceed 1^0 in a length of 20 times the shaft diameter. Design and draw a Protected type flange coupling (Coupling bolt: allowable shear stress = 30 MPa, key material – mild steel, modulus of rigidity of mild steel is 84×10^3 MPa.).

OR

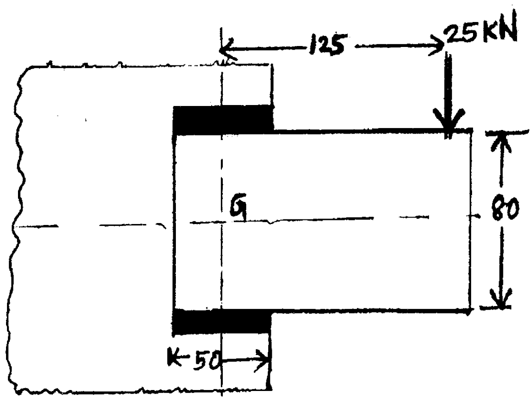
V. Design and draw a cotter joint to connect two rods, which transmit maximum tensile load of 50 kN. Assume the rods, sleeve and cotter are made of steel, permissible stress in the material are 55 MPa in tension and 70 MPa in crushing and 35 MPa in shear.

VI. Design the riveted joints for the longitudinal seam for a 1.5 m diameter steam boiler to carry a steam pressure of 1 MPa. The ultimate strength of the boiler plate may be assumed as 300 MPa, crushing strength for rivets as 600 MPa and shear strength for rivets as 300 MPa. Take the joint efficiency as 80% and assume factor of safety as 5. Sketch the joint with all dimensions.

OR

VII. An engine indicator has a plunger diameter of 20 mm. When the steam pressure on the plunger is 5 MPa, the indicator spring is compressed by 12 mm. For permissible shear stress of 450 MPa, spring index of 5 and modulus of rigidity 80×10^3 MPa, design the helical compression spring.

VIII. A bracket carrying a load of 25 kN is to be welded as shown in figure. Find the size of weld required if the allowable shear stress is not to exceed 90 MPa.



(All dimensions in mm)

OR

IX. A hollow shaft of 0.5 m outside diameter and 0.3 m inside diameter is used to drive a propeller of a marine vessel. The shaft is mounted on bearings 6 m apart and it transmits 5600 kW at 1500 rpm. The maximum axial propeller thrust is 500 kN and the shaft weighs 70 kN.

Determine (i) The maximum shear stress developed in the shaft.
(ii) The angular twist between the bearings.

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